

THE UNIVERSITY OF HONG KONG
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

COMP7103 Data mining

Date: May 12th 2022

Time: 6:30pm - 8:30pm

There are five questions in total, the mark value of the question (or part of the question) is indicated before the question (or part of the question). All questions are compulsory.

This is a closed book examination. Internet searching and crowdsourcing from group messages, online forums or social media, etc. are strictly forbidden.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Candidates should provide handwritten answers on self-prepared blank paper.
Candidates should submit a single PDF file to OLEX.

Question 1 (Naive Bayes) 20 pts. Given the training data in Table 1, where “Flu” is the class,

- a) (5 pts) use the Naive-Bayes classifier to predict the class of (Running Nose = ‘Y’, Coughing = ‘Y’, Fever=‘Y’).
- b) (5 pts) Assume data points d_1 and d_2 contain the following missing values: for d_1 we do not know whether the patient has fever, while for d_2 we do not know whether the patient has a running nose. In other words, d_1 becomes $d_1=(N='Y',C='Y',F='?',Flu='Y')$, while d_2 becomes $d_2=(N='?',C='Y',F='N',Flu='N')$. Predict the class of (Running Nose = ‘Y’, Coughing = ‘Y’, Fever=‘Y’) after removing any data point with a missing value;
- c) (5 pts) consider the same data in b). Predict the class of (Running Nose = ‘Y’, Coughing = ‘Y’, Fever=‘Y’) as follows: when computing conditional probabilities for any feature that contains missing values, ‘pretend’ to remove data points with missing values for that feature.
- d) (5 pts) Is it necessary to compute Prob (N = ‘Y’, C = ‘Y’, F=‘Y’) in any of the questions above? In the positive case, argue why it is important to compute such a quantity. Otherwise, argue why it is important not to compute such a quantity.

Training Example	Running nose (N)	Coughing (C)	Fever (F)	Flu
d_1	Y	Y	Y	Y
d_2	Y	Y	N	N
d_3	N	N	N	N
d_4	Y	Y	N	N
d_5	Y	Y	N	Y
d_6	N	Y	Y	Y
d_7	Y	N	N	N

Table 1.

Question 2 (Decision Trees) 20 pts.

- a) (5 pts) Build a decision tree for the data reported in Table 2. You should use the following rules: i) at each step compute the gini index for every possible split and select the split with the best gini value. (ii) stopping rule: when the gini value of a node is zero or no further split is possible; (iii) the class of a leaf node is determined by the majority rule with 0 being the default class. Show all the steps.
- b) (5 pts) After building the tree, prune the tree (if needed) so as to improve the generalization error=training error + $0.5 \times$ number of leaves. Show all the steps. Explain the criteria used to stop the pruning.
- c) (5 pts) Is the tree you obtained after pruning (if pruned) the one that minimizes the generalization error? Explain your answer.
- d) (5 pts) Predict the class of (A=1,B=1,C=1) using the tree after pruning.

A	B	C	Class
0	0	1	0
0	0	1	0
1	0	1	1
1	1	1	1
0	1	1	0
0	2	1	1
0	2	1	1
0	2	0	1
0	2	0	0

Table 2.

Question 3 (Pagerank) 20 pts.

- a) (10 pts) Compute the PageRank vector for the graph in Figure 1 (without random jumps).
- b) (10 pts) Run the PageRank algorithm to compute the first two vectors starting from $(1/5, 1/5, 1/5, 1/5, 1/5)$ (without random jumps).

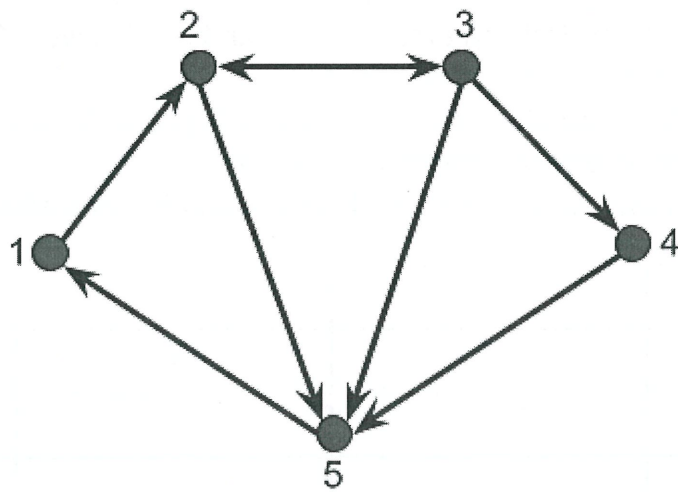


Figure 1.

Question 4 (k-means) (20 pts).

a) (10pts) Consider the following points in the Euclidean space: $P_1=(0,0)$, $P_2=(1,0)$, $P_3=(3,0)$, $P_4=(4,0)$, $P_5=(1,1)$, $P_6=(4,1/4)$, $P_7=(2,2)$, $P_8=(2,3)$, $P_9=(3,3)$. Run the k-means algorithm until convergence, starting with P_1, P_2 , and P_5 as initial centroids ($k=3$). Show all the steps, as well as the clusters obtained at the end of each step.

b) (5pts) Choose the initial centroids so that k-means terminates in one iteration and show that indeed the algorithm terminates in one iteration.

c) (5pts) Does k-means always terminate for any possible choice of the initial centroids?

Question 5. (Densest Subgraph) (20 pts). Consider the graph G represented in Figure 2.. Recall the greedy algorithm for computing a dense subgraph: at each step one node with minimum degree is removed; at the end we output the subgraph $H=(V_H, E_H)$ with maximum density = $|E_H|/|V_H|$ among all considered subgraphs.

- (6pts) Run the greedy algorithm on G and report the subgraph computed by such an algorithm. If there are two nodes with the same degree, you should remove the one with a smaller index. E.g. between 8 and 9 you should remove 8.
- (6pts) What is the densest subgraph in G (i.e. it maximizes $|E_H|/|V_H|$)?
- (8pts) Change the first node to be removed so that greedy computes a densest subgraph. You can only choose the first node to be deleted, the rest of the algorithm is unchanged. This applies only in the case when the densest subgraph was not found already in a).

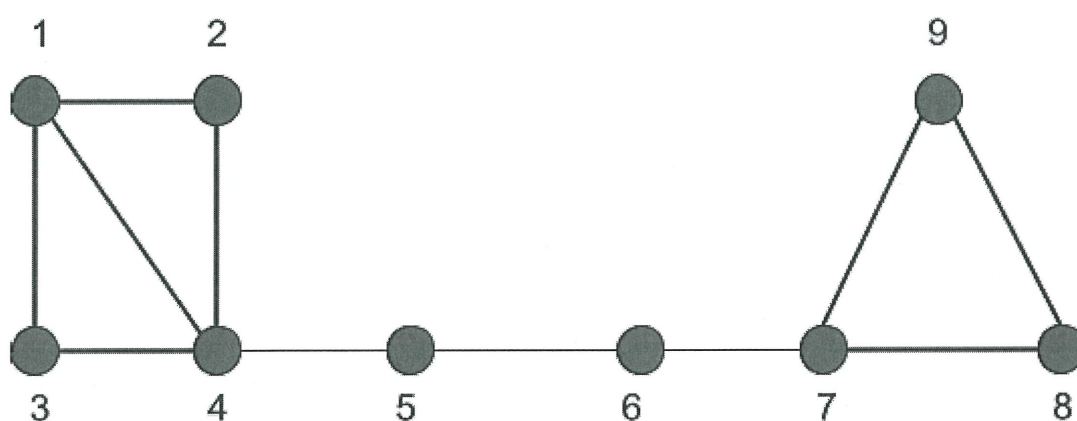


Figure 2.

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